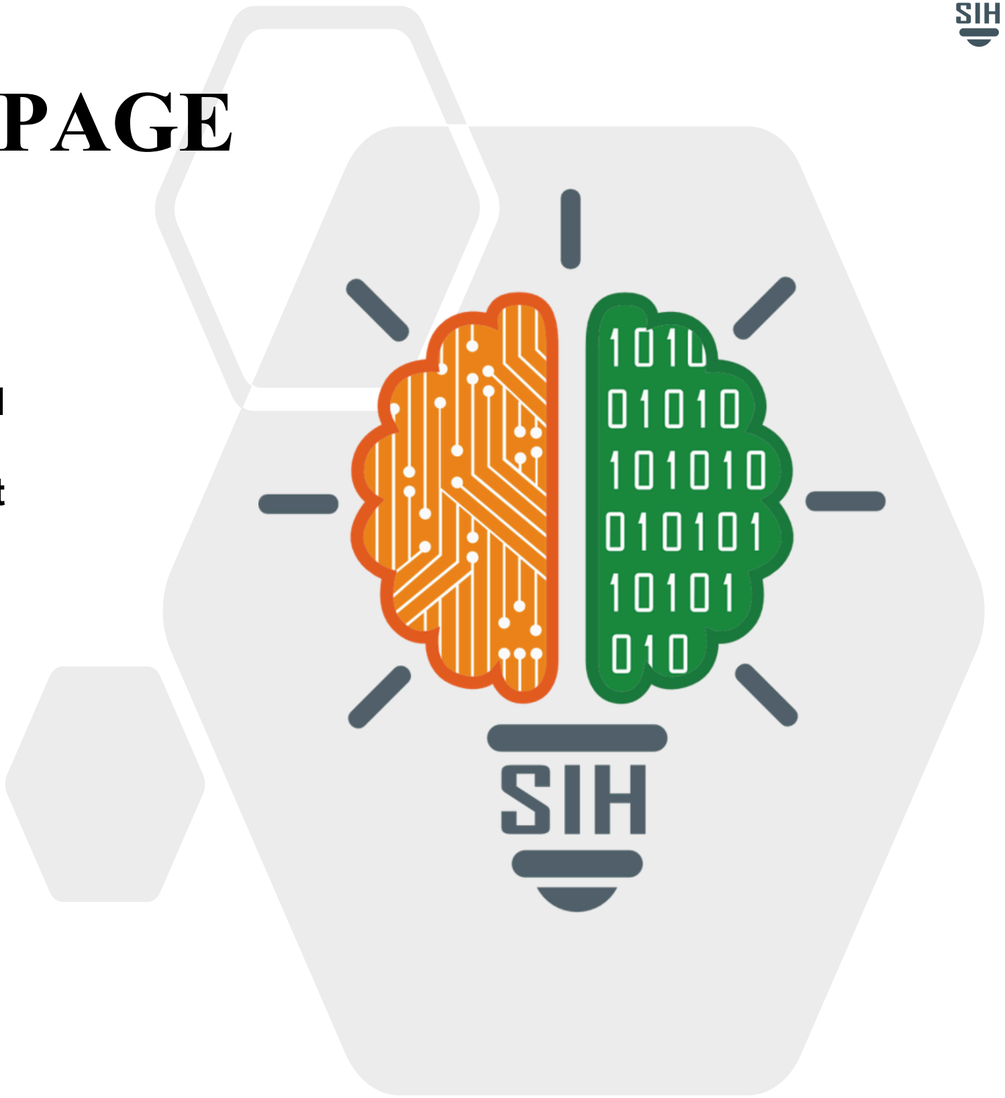
TITLE PAGE

- **Problem Statement ID –26262**
- **Problem Statement Title-Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 198299**
- **DEEP DRISHTI**



DEEP DRISHTI

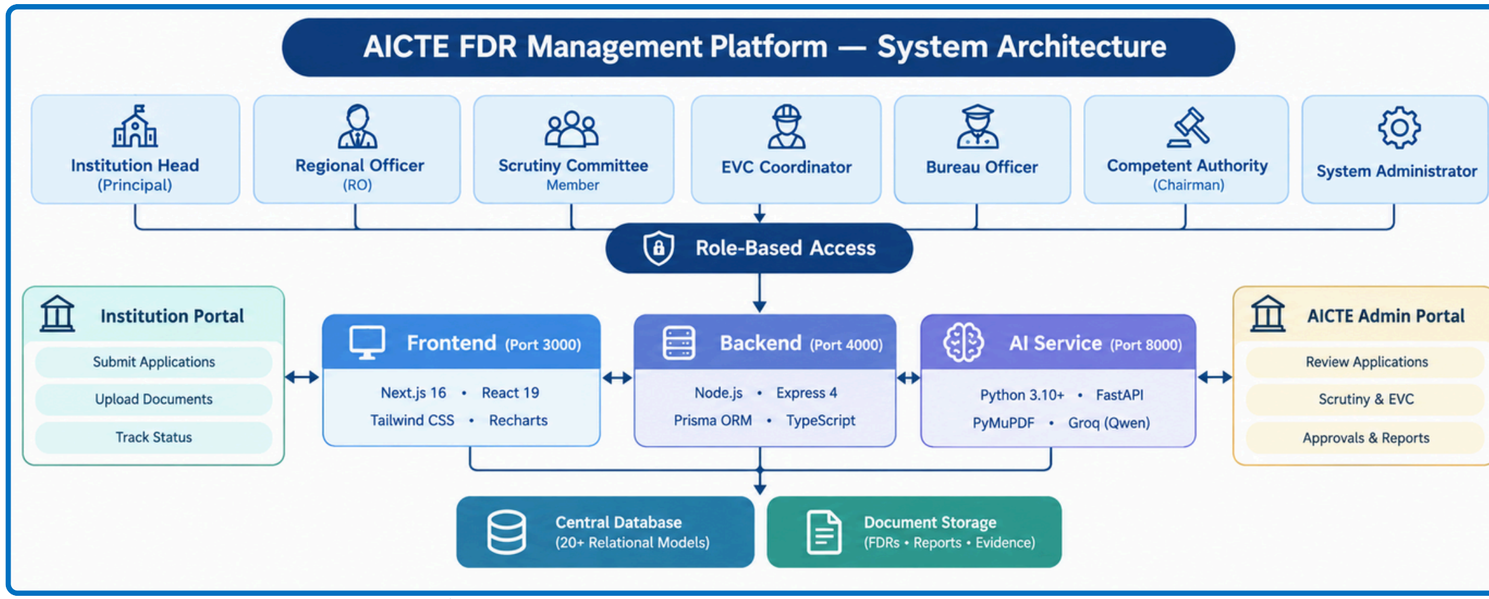
IDEA TITLE

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Proposed Solution

An AI-powered, end-to-end digital ecosystem designed to automate and streamline AICTE's institutional approval process.

- **Self-Service Institution Portal:** A 5-step guided filing wizard with dynamic category checklists, a statutory document vault, and a real-time deficiency resolution desk for Annual Extension of Approval (EoA), new campuses, and intake variations.
- **Command & Control AICTE Official Portal:** A central triage queue, granular scrutiny consoles, automated EVC inspection scheduling, and conditional approval management for AICTE officials.
- **Autonomous Citizen's Charter SLA Engine:** Real-time countdown timers on every statutory stage gate, with automated 3-tier escalation notices (Regional Officer → Bureau Officer → Chairman).
- **Deterministic + Agentic AI Pre-Scrutiny:** PyMuPDF document extraction plus Groq-hosted LLM (Qwen 3.8 27B) advisory checks that pre-screen documents against APH 2024–2027 norms before human scrutiny.
- **Closed-Loop Post-Approval Governance:** Conditional approvals automatically spawn actionable compliance undertakings, with evidence upload by the institution and verification by AICTE officials.
- **Role-Based Workflow Automation:** Seven strictly segregated statutory personas, from Institution Head to Chairman, each with scoped visibility and permissions.



Why we stand out?

Hybrid AI Scrutiny Engine:

Combines statutory rule-based validation and LLM reasoning to detect regulatory violations and cross-document inconsistencies.

Human-in-the-Loop Advisory:

Provides advisory AI findings with compliance badges and APH clause references while keeping final decisions under officer authority.

Full-Lifecycle Regulatory State Machine:

Automates statutory workflows from Filing to Final Order with secure, controlled stage transitions.

Active 3-Tier SLA Escalation

Automatically escalates overdue cases after 7, 14 and 21 days to designated authorities.

Tamper-Evident Audit Trail:

Uses SHA-256 fingerprinting and append-only logs to ensure document integrity and trace every action and decision.

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TECHNICAL APPROACH

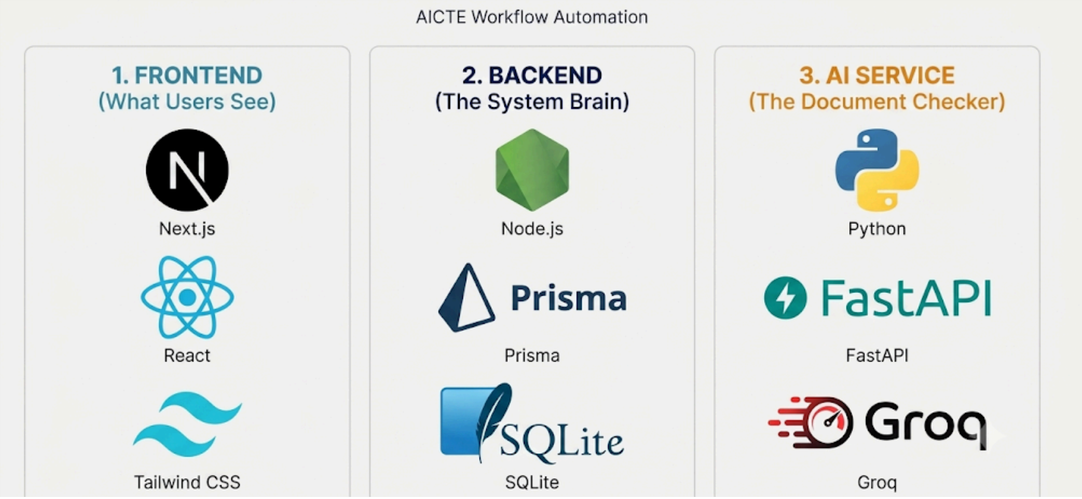
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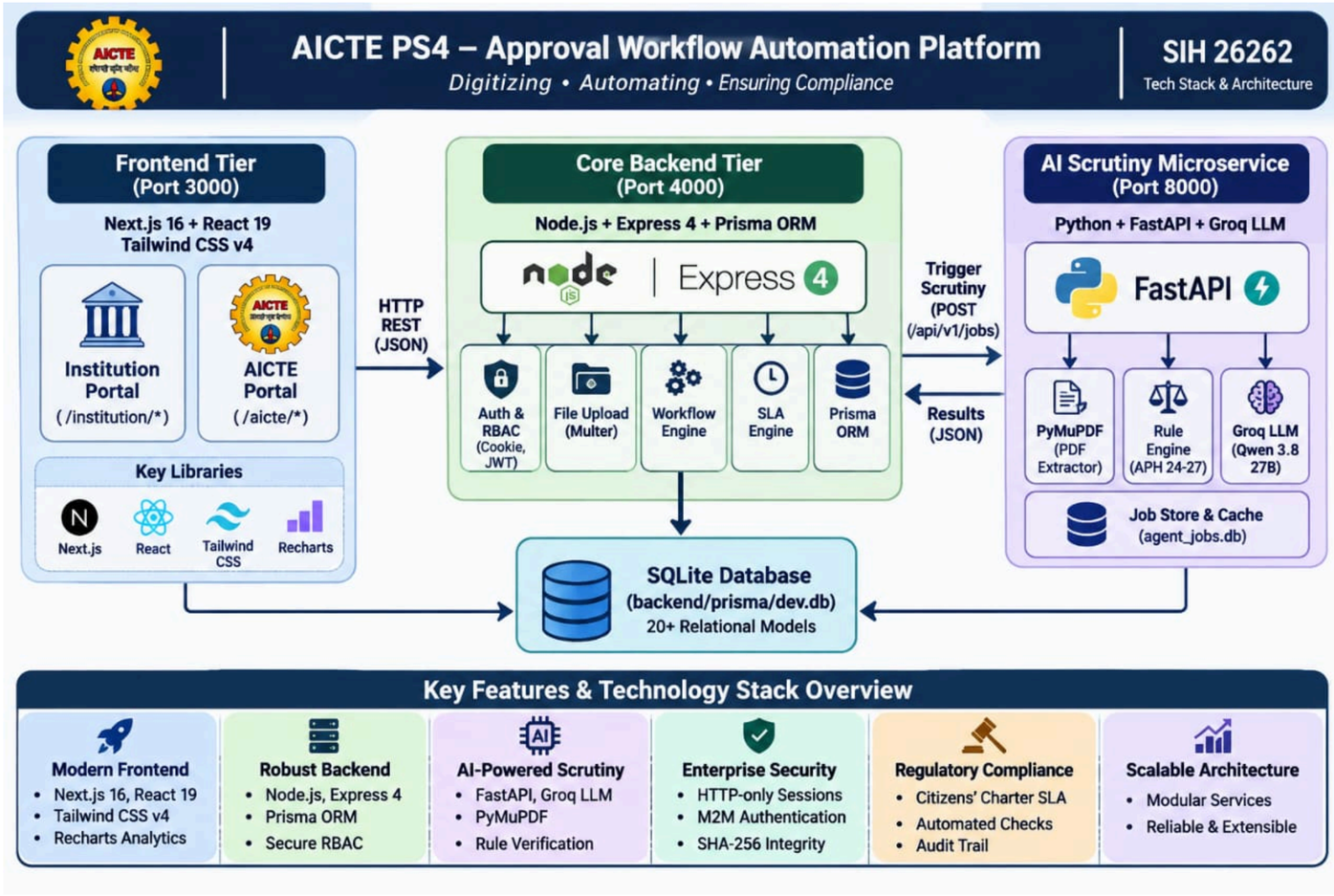
Technology Stack

Frontend (Port 3000): Next.js 16,React 19,Tailwind CSS v4,Turbopack,Recharts,Lucide React

Backend (Port 4000)
- Node.js, Express 4, TypeScript, Prisma ORM 6, Zod, Multer, SQLite (Development)

AI Service (Port 8000)
- Python 3.10+, FastAPI, Uvicorn, PyMuPDF, Groq (Qwen 3.8 27B)





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FEASIBILITY AND VIABILITY

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Technical Feasibility



- Next.js 16, React 19 and Tailwind CSS v4 frontend.
- Node.js/Express 4 with TypeScript backend.
- Prisma ORM 6 with 20+ relational models.
- Python FastAPI with PyMuPDF and Groq AI inference.

Security Feasibility



- HTTP-only session cookies.
- Bcryptjs password hashing.
- Zod-based payload validation.
- Strict role-based access control (RBAC).
- SHA-256 document hashing and append-only audit logs.

Integration Feasibility



- RESTful API-based architecture.
- DigiLocker/NAD integration for verified documents.
- Bhashini integration for multilingual access.
- Extensible third-party service connectivity.

Potential Challenge & Risks

- **AI Errors:** Incorrect or hallucinated AI findings.
- **Service Latency:** AI delays or API rate limits.
- **Document Quality:** Blurred, incomplete or inconsistent uploads.
- **Officer Backlog:** Delays in scrutiny and approvals.
- **Security Threats:** Unauthorized access or data tampering.
- **Compliance Gaps:** Unfulfilled post-approval conditions.
- **Scalability:** Increasing data and storage requirements.

Strategies To Overcome Challenges

- **AI Errors:** Human officer verification and rule-based statutory validation.
- **Service Latency:** Isolated FastAPI service with backend fallback.
- **Document Quality:** Low-confidence flagging, manual review and 10-day resubmission window.
- **Officer Backlog:** SLA countdowns and 3-tier escalation system.
- **Security Threats:** HTTP-only cookies, RBAC, SHA-256 hashing and audit logs.
- **Compliance Gaps:** Automated undertakings, deadlines and evidence verification.
- **Scalability:** PostgreSQL migration, cloud storage and Kubernetes deployment.
- **Low Adoption:** Guided onboarding, dynamic checklists and simple status tracking.

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IMPACT AND BENEFITS

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Faster Approval Cycles

Average processing time of 29.7 days against the 45-day benchmark, about 34% faster.

High Citizen's Charter SLA Compliance

A 96.4% on-time resolution rate, backed by automated escalations.



Reduced Manual Review Effort

AI pre-screening reduces review effort by over 60%, so scrutiny committees can focus on high-judgment verification.

Continuous Post-Approval Compliance

Conditions such as fire NOC renewal or land conversion are tracked until certified compliant.

100% Real-Time Transparency

Institutions track their application live and no longer depend on opaque, letter-based communication.

Benefits Of the Solution

Social

- Equal Accessibility: Centralised access for institutions across all regions.
- Faster Communication: Digital deficiency alerts and simplified responses.
- Stakeholder Trust: Traceable decisions and accountable workflows.

Economic

- Reduced Operational Costs: Minimises printing, courier and physical record expenses.
- Resource Optimisation: Reduces repetitive manual verification.
- Lower Administrative Overheads: Automates alerts, tracking and reporting.

Environmental

- Lower Paper Consumption: Minimises printing and photocopying.
- Reduced Transportation: Limits physical document movement between offices.
- Digital Record Management: Reduces physical storage requirements.
- Lower Carbon Footprint: Minimises resource consumption through digital workflows.

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RESEARCH AND REFERENCES

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Feature	Current AICTE Portal	Our Automated Solution (Your automated, AI-driven website)
Application Routing & Workflow	Linear, rigid manual reviews (no bypass routes or formal escalation)	Stateful transition engine with automated workflows (Re-Scrutiny & SAC loops)
Document Scrutiny	Manual, visual verification of multi-page paper/PDF uploads (slow)	Asynchronous OCR with Groq-powered AI (Qwen 3.8 27B) checking and highlighting errors
Deadlines & Enforcement	Manual email tracking with high risk of invisible file bottlenecks	Citizen's Charter SLA engine with automatic tier escalations to the Chairman
Audit Trails & Accountability	Logins with unmonitored updates (susceptible to back-channel edits)	Cryptographic SHA-256 and immutable, append-only AuditEvent log
Access Control	Basic username/password login for users	7 Role-Based persona limits with Aadhaar/PAN signature checks

https://cdnbbsr.s3waas.gov.in/s35938b4d054136e5d59ada6ec9c295d7a/uploads/2025/03/2025031399.pdf

https://aicte.gov.in/sites/default/files/Tender-SI-Final%20RFP%20after%20adding%20extension%20clause.pdf

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.